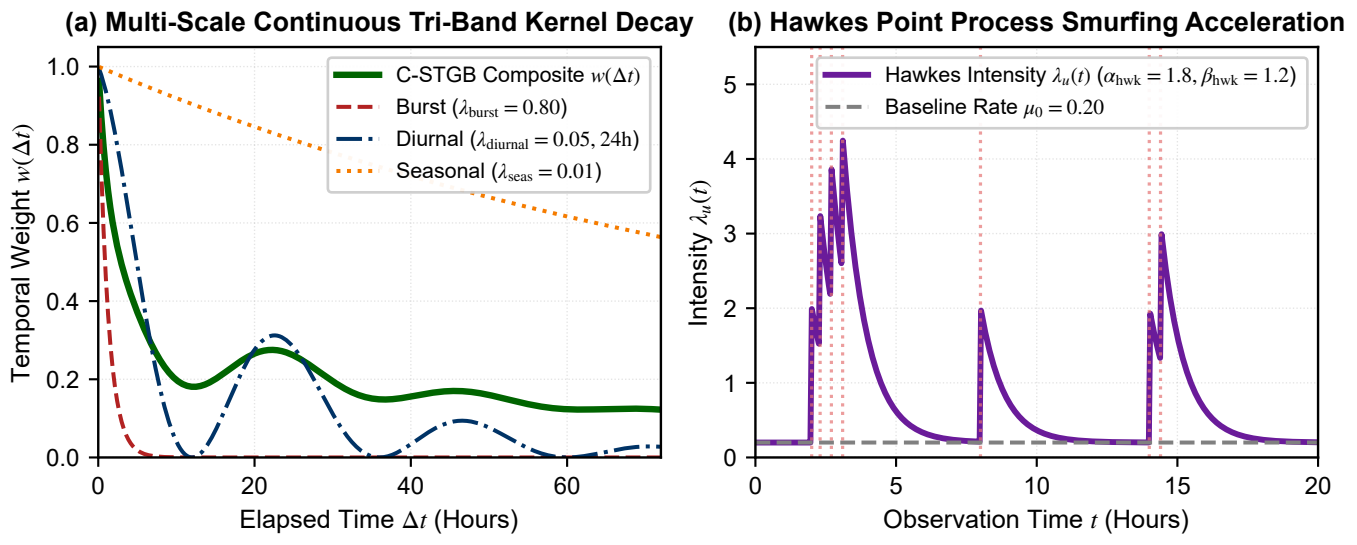

(a) Multi-Scale Continuous Tri-Band Kernel Decay
C-STGB Composite $w(\Delta t)$
Burst ($\lambda_{\text{burst}} = 0.80$)
Diurnal ($\lambda_{\text{diurnal}} = 0.05$, 24h)
Seasonal ($\lambda_{\text{seas}} = 0.01$)
Temporal Weight $w(\Delta t)$
Elapsed Time Δt (Hours)
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
(b) Hawkes Point Process Smurfing Acceleration
Hawkes Intensity $\lambda_u(t)$ ($\alpha_{\text{hwk}} = 1.8$, $\beta_{\text{hwk}} = 1.2$)
Baseline Rate $\mu_0 = 0.20$
Intensity $\lambda_u(t)$
Observation Time t (Hours)
0
1
2
3
4
5
0
5
10
15
20